



TOOLS | AUGUST 28, 2009

Nmap Cheat Sheet

Nmap has a multitude of options, when you first start playing with this excellent tool, it can be a bit daunting.

In this cheat sheet, you will find a series of practical example commands for running Nmap and getting the most of this powerful tool.

Keep in mind this cheat sheet merely touches the surface of the available options. The [Nmap Documentation portal](#) is your reference for digging deeper into the options available.



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HACKER TARGET

- Target Selection
- Port Selection
- Scan Types
- Service and Operating System Detection
- Output formats
- Nmap CSV Output
- NSE Scripting
- Find DDOS reflectors
- HTTP info gathering
- Heartbleed Detection
- IP Information Gathering
- Remote Scanning
- Additional Resources

Nmap Target Selection

Scan a single IP	<code>nmap 192.168.1.1</code>
Scan a host	<code>nmap www.testhostname.com</code>
Scan a range of IPs	<code>nmap 192.168.1.1-20</code>
Scan a subnet	<code>nmap 192.168.1.0/24</code>
Scan targets from a text file	<code>nmap -iL list-of-ips.txt</code>

These are all default scans, which will scan 1000 TCP ports. Host discovery will take place.

Nmap Port Selection

Scan a single Port	<code>nmap -p 22 192.168.1.1</code>
Scan a range of ports	<code>nmap -p 1-100 192.168.1.1</code>
Scan 100 most common ports (Fast)	<code>nmap -F 192.168.1.1</code>
Scan all 65535 ports	<code>nmap -p- 192.168.1.1</code>

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Scan UDP ports	<code>nmap -sU -p 123,161,162 192.168.1.1</code>
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Scan selected ports - ignore discovery	<code>nmap -Pn -F 192.168.1.1</code>
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Privileged access is required to perform the default [SYN](#) scans. If privileges are insufficient a TCP connect scan will be used. A TCP connect requires a full TCP connection to be established and therefore is a slower scan. Ignoring discovery is often required as many firewalls or hosts will not respond to [PING](#), so could be missed unless you select the `-Pn` parameter. Of course this can make scan times much longer as you could end up sending scan probes to hosts that are not there.

Take a look at the [Nmap Tutorial](#) for a detailed look at the scan process.

Service and OS Detection

Detect OS and Services	<code>nmap -A 192.168.1.1</code>
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Standard service detection	<code>nmap -sV 192.168.1.1</code>
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More aggressive Service Detection	<code>nmap -sV --version-intensity 5 192.168.1.1</code>
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Lighter banner grabbing detection	<code>nmap -sV --version-intensity 0 192.168.1.1</code>
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Service and OS detection rely on different methods to determine the operating system or service running on a particular port. The more aggressive service detection is often helpful if there are services running on unusual ports. On the other hand the lighter version of the service will be much faster as it does not really attempt to detect the service simply grabbing the banner of the open service.

Nmap Output Formats

Save default output to file	<code>nmap -oN outputfile.txt 192.168.1.1</code>
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Save results as XML	<code>nmap -oX outputfile.xml 192.168.1.1</code>
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Save results in a format for grep	<code>nmap -oG outputfile.txt 192.168.1.1</code>
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Save in all formats	<code>nmap -oA outputfile 192.168.1.1</code>
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Nmap by default has no [csv](#) output format. Use the [XML](#) output to extract the relevant fields into [csv](#) with [python](#).

Jump over to [github](#) and grab our sample script that can be easily modified depending on your requirements. With [csv](#) files it is easy to convert into [xlsx](#) for reporting. This can be done manually or using our [python](#) conversion script.

Nmap XML to CSV

Digging deeper with NSE Scripts

Scan using default safe scripts

```
nmap -sV -sC 192.168.1.1
```

Get help for a script

```
nmap --script-help=ssl-heartbleed
```

Scan using a specific NSE script

```
nmap -sV -p 443 --script=ssl-heartbleed.nse  
192.168.1.1
```

Scan with a set of scripts

```
nmap -sV --script=smb* 192.168.1.1
```

According to my Nmap install there are currently **581 NSE scripts**. The scripts are able to perform a wide range of security related testing and discovery functions. If you are serious about your network scanning you really should take the time to get familiar with some of them.

The option `--script-help=$scriptname` will display help for the individual scripts. To get an easy list of the installed scripts try `locate nse | grep script`.

You will notice I have used the `-sV` service detection parameter. Generally most NSE scripts will be more effective and you will get better coverage by including service detection.

A scan to search for DDOS reflection UDP services

Scan for UDP DDOS reflectors

```
nmap -sU -A -PN -n -pU:19,53,123,161 --script=ntp-monlist,dns-recursion,snmp-sysdescr 192.168.1.0/24
```

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Gather page titles from HTTP services	<code>nmap --script=http-title 192.168.1.0/24</code>
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Get HTTP headers of web services	<code>nmap --script=http-headers 192.168.1.0/24</code>
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Find web apps from known paths	<code>nmap --script=http-enum 192.168.1.0/24</code>
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There are many HTTP information gathering scripts, here are a few that are simple but helpful when examining larger networks. Helps in quickly identifying what the HTTP service that is running on the open port. Note the [http-enum](#) script is particularly noisy. It is similar to [Nikto](#) in that it will attempt to enumerate known paths of web applications and scripts. This will inevitably generated hundreds of [404 HTTP responses](#) in the web server error and access logs.

Detect Heartbleed SSL Vulnerability

Heartbleed Testing	<code>nmap -sV -p 443 --script=ssl-heartbleed 192.168.1.0/24</code>
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Heartbleed detection is one of the available SSL scripts. It will detect the presence of the well known Heartbleed vulnerability in SSL services. Specify alternative ports to test SSL on mail and other protocols (*Requires Nmap 6.46*).

IP Address information

Find Information about IP address	<code>nmap --script=asn-query,whois,ip-geolocation-maxmind 192.168.1.0/24</code>
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Gather information related to the IP address and netblock owner of the IP address. Uses ASN, whois and geoip location lookups. See the [IP Tools](#) for more information and similar IP address and DNS lookups.

Remote Scanning

Testing your network perimeter from an external perspective is key when you wish to get the most accurate results. By assessing your exposure from the attackers perspective you can validate firewall rule audits and understand exactly what is allowed into your network. This is the reason we offer a

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The above commands are just a taste of the power of Nmap. Check out our [Nmap Tutorial](#) that has more information and tips.

You could also view the full set of features by running Nmap with no options. The creator of Nmap, Fyodor, has a [book](#) available that covers the tool in depth.

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